

Curriculum Vitae of the Candidate
Programa Doutoral do Centro Académico Médico de Lisboa (CAML)

1. Full Name: Alexandre Miguel Kaizeler Guedes da Silva Afonso

2. Birth Date: 5th of September, 1997

3. Address: Rua Aquilino Ribeiro, R/C Dto, N°14 2675-294 Odivelas, Portugal

4. Telephone/Mobile: +351912140319

5. E-mail: alexandre.afonso@medicina.ulisboa.pt

6. Academic and Professional Degrees:

Year	Degree	Degree granting institution	Final grade
2021	MSc in Oncobiology	Universidade de Lisboa	18
2018	BSc in Biomedical Sciences	Universidade do Algarve	17

7. Current Status:

Starting date	Position/Job	Institution
January 2022	PhD student	Faculdade de Medicina da Universidade de Lisboa Instituto de Medicina Molecular João Lobo Antunes (iMM) / Gulbenkian Institute for Molecular Medicine (GIMM)

8. Previous Activities:

Period	Position/Job	Institution
September 2019 – December 2020	Master Thesis Project	Fundação Champalimaud

9. Present Research Interests:

Interested in transcriptomics and computational biology applied to health and disease, in particular in relation to cancer cell biology.

10. Area of Scientific Activity

10.1. Current: Transcriptomic analyses regarding stress granules' transcriptomes, and transcripts of stress granules-harboring cells. Identification and correction of methodological biases identified in stress granule transcriptomes and development of methods to infer stress granule presence in human cancer transcriptomes and relate these to patient clinical data.

10.2. Previous: Master's thesis project in cancer cell biology, applied to cell competition, identifying phenotypical and transcriptome changes in both competing cells.

11. Participation in R&D projects:

Project Title	Funding Entity	Total Budget	Role ^a
BIOMICS Fostering Excellent Research, Training and Innovation in Biomedical Data Science	European Union	€ 1 493 729,38	Research Fellow

12. Published Work:

12.1. Papers in Peer-Reviewed Journals

- Martins-Silva R, **Kaizeler A**, Barbosa-Morais NL (2026). Exploring molecular signatures of senescence with markeR, an R toolkit for evaluating gene sets as phenotypic markers. NAR Genomics and Bioinformatics.
<https://doi.org/10.1093/nargab/lqag057>
- Wagner, V., Bousset, L., Ascensão-Ferreira, M. et al. (2026). Spliceosome induction is a druggable dependency of RAS-driven senescence and cancer. Nat Commun. <https://doi.org/10.1038/s41467-026-71564-z>
- Schneider AL*, Martins-Silva R*, **Kaizeler A***, Saraiva-Agostinho N, Barbosa-Morais NL (2024). voyAGER: free web interface for the analysis of age-related gene expression alterations in human tissues. eLife 12:RP88623.
<https://doi.org/10.7554/eLife.88623.3>
- Camacho D, Jesus JP, et al. (2020) SPARC-p53: The double agents of cancer. Advances in cancer research, 148, 171-199.
<https://doi.org/10.1016/bs.acr.2020.05.004>

* these authors contributed equally to the work

13. Scientific Communications (Platform Presentations, Posters, Abstracts)

- One-minute Pitch Presentation | Kaizeler, Alexandre; Barbosa-Morais, Nuno. “CHARM: A Comprehensive Hub for Alternative Regulatory Mapping of RNA-Binding Proteins”. Bioinformatics Open Days, Universidade do Minho, Campus Gualtar, Braga, Portugal. 25 March 2026.
- Software Poster Presentation | Kaizeler, Alexandre; Barbosa-Morais, Nuno. “CHARM: A Comprehensive Hub for Alternative Regulatory Mapping of RNA-Binding Proteins”. Bioinformatics Open Days, Universidade do Minho, Campus Gualtar, Braga, Portugal. 25 March 2026.
- Oral Presentation | Kaizeler, Alexandre; Barbosa-Morais, Nuno. “Revisiting Stress Granule Transcriptomes Reveals Mitochondrial RNA Enrichment Despite Methodological Bias”. 1st CAML & GIMM PhD meeting, Gulbenkian Institute for Molecular Medicine, Faculdade de Medicina da Universidade de Lisboa, Lisbon, Portugal. 26 January 2026.
- Oral Presentation | Kaizeler, Alexandre; Barbosa-Morais, Nuno. “CHARM: A Comprehensive Hub for Alternative Regulatory Mapping of RNA-Binding Proteins”. RNA Mini Symposium, Gulbenkian Institute for Molecular Medicine, Faculdade de Medicina da Universidade de Lisboa, Lisbon, Portugal. 23 January 2026.
- Poster Presentation | Kaizeler, Alexandre; Barbosa-Morais, Nuno. “Deciphering the Role of Stress Granules: Dampening Inflammation in Cancer through

- Mitochondrial RNA Sequestration.” EACR 2025 Congress: Innovative Cancer Science, Lisbon (Portugal), 17 June 2025.
- Oral Presentation | Kaizeler, Alexandre; Barbosa-Morais, Nuno. “Stress Granules: A Novel Biomarker for Cancer Prognosis? A Machine Learning Approach” Oncobiology Seminars, Gulbenkian Institute for Molecular Medicine, Faculdade de Medicina da Universidade de Lisboa, Lisbon, Portugal. 16 April 2025
 - Oral Presentation | Kaizeler, Alexandre; Barbosa-Morais, Nuno. “Triple Entente: Mitochondrial Leakage, Stress Granules and the War on Inflammation” RNA Club Seminars, Gulbenkian Institute for Molecular Medicine, Faculdade de Medicina da Universidade de Lisboa, Lisbon, Portugal, 17 January 2025
 - Oral Presentation | Kaizeler, Alexandre; Barbosa-Morais, Nuno. “The Role of RNA Length in the Stress Granule Transcriptome” Computational Biology and Bioinformatics Seminars (CBBS), Instituto de Medicina Molecular João Lobo Antunes, Faculdade de Medicina da Universidade de Lisboa, Lisbon, Portugal. 26 June 2024.
 - Oral Presentation | Kaizeler, Alexandre; Barbosa-Morais, Nuno. “The Role of RNA Length in the Stress Granule Transcriptome” Pizza Seminars, Instituto de Medicina Molecular João Lobo Antunes, Faculdade de Medicina da Universidade de Lisboa, Lisbon, Portugal. 4 June 2024.
 - Poster Presentation | Kaizeler, Alexandre; Barbosa-Morais, Nuno. “The Role of RNA Length in the Stress Granule Transcriptome” XVII CAML PhD Students meeting, Instituto de Medicina Molecular João Lobo Antunes, Faculdade de Medicina da Universidade de Lisboa, Lisbon, Portugal. May 2024.
 - Oral Presentation | Kaizeler, Alexandre; Barbosa-Morais, Nuno. “The Role of RNA Length in the Stress Granule Transcriptome” RNA Club, Instituto de Medicina Molecular João Lobo Antunes, Faculdade de Medicina da Universidade de Lisboa, Lisbon, Portugal. 19 January 2024.
 - Poster Presentation | Kaizeler, Alexandre; Barbosa-Morais, Nuno. “Does Size Matter? The Role of RNA Length in the Stress Granule Transcriptome” EMBO Young Scientists’ Forum, Instituto de Medicina Molecular João Lobo Antunes, Faculdade de Medicina da Universidade de Lisboa, Lisbon, Portugal. 12 October, 2023.
 - Poster Presentation | Kaizeler, Alexandre; Barbosa-Morais, Nuno. “A Computational Approach to Define a Stress Granule Signature and its Implication in Cancer” XVI CAML PhD Students Meeting, Instituto de Medicina Molecular João Lobo Antunes, Faculdade de Medicina da Universidade de Lisboa, Lisbon, Portugal. 11 May, 2023.
 - Oral Presentation | Kaizeler, Alexandre; Queirós, Ana. “Cell Competition: Separating the Winners from the Losers” Oncobiology Seminars, Instituto de Medicina Molecular João Lobo Antunes, Faculdade de Medicina da Universidade de Lisboa, Lisbon, Portugal. 15 March, 2023.
 - Poster Presentation | Kaizeler, Alexandre; Barbosa-Morais, Nuno. “A Computational Approach to Define a Stress Granule Signature and its

Implication in Cancer” RiboMed and IX ptRNA joint meeting, Instituto de Medicina Molecular João Lobo Antunes, Faculdade de Medicina da Universidade de Lisboa, Lisbon, Portugal. 26 January, 2023.

- Oral Presentation | Kaizeler, Alexandre, Queirós, Ana. “Cell Competition: Separating the Winners from the Losers” Computational Biology and Bioinformatics Seminars (CBBS), Instituto de Medicina Molecular João Lobo Antunes, Faculdade de Medicina da Universidade de Lisboa, Lisbon, Portugal. 25 January, 2023.
- Poster Presentation | Afonso, Alexandre, Barbosa-Morais, Nuno. “A Computational Approach to Define a Stress Granule Signature and its Implication in Cancer” EMBO | EMBL Symposium The complex life of RNA 2022, EMBL, Heidelberg, Germany. 13 October, 2022.

14. Scholarships

- Research Fellowship, February 2026 – July 2026, Gulbenkian Institute for Molecular Medicine (BI-70-2025)
- Research Fellowship, February 2022 – January 2026, NMorais Lab, FCT (UI/BD/153363/2022)